Task 15: Final End to End Release Documentation and Viva

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1. Objective

Document the final end-to-end DevOps release and provide concise viva preparation based on the implementation actually completed.

2. Final Workflow

```
Developer
  |
Git / GitHub
  |
Jenkins
  |
Maven Build + Automated Tests
  |
Docker Image
  |
Docker Container
  |
Host :8766 -> Container :8765
  |
Puppet-managed configuration
  |
Verified release
```

3. Final Technology Stack

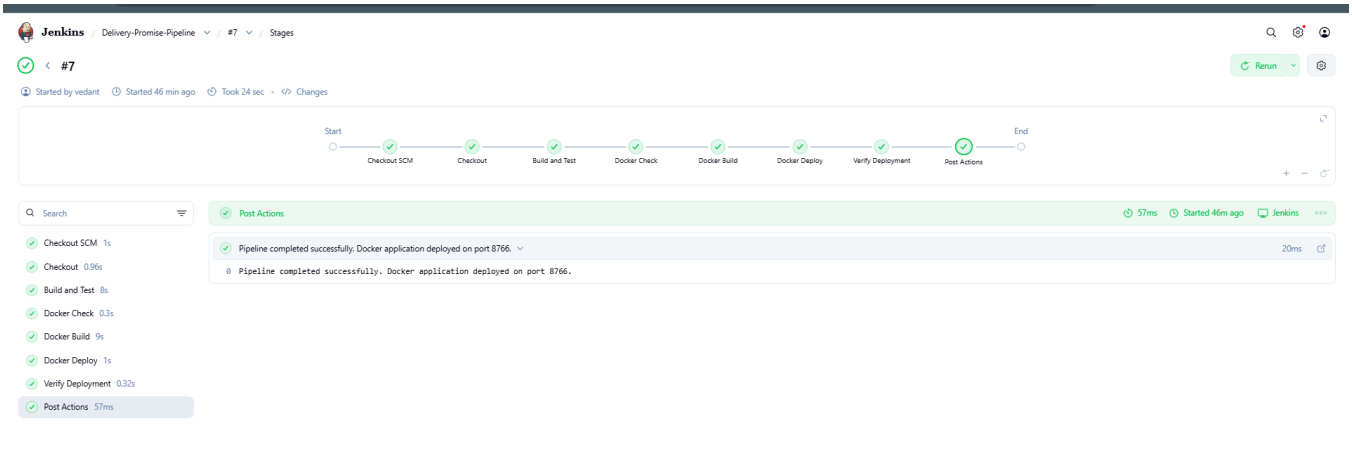
Component	Project technology
Language/Runtime	Java 21
Framework	Spring Boot 4.1.1
Build	Maven 3.9.16
Database	SQLite + Spring Data JPA/Hibernate
UI	Thymeleaf + HTML/CSS
CI/CD	Jenkins
Testing	Selenium WebDriver + JUnit 5
Containerization	Docker
Configuration	Puppet 8.10.0
Version control	Git + GitHub

4. Final Jenkins Pipeline

The final pipeline includes Checkout SCM, Checkout, Build and Test, Docker Check, Docker Build, Docker Deploy, Verify Deployment, and Post Actions.

5. Final Evidence

Evidence: final successful Jenkins pipeline



6. Final Release Result

Jenkins build #7 completed successfully in 24 seconds. Every displayed stage passed and the post action confirmed Docker deployment on port 8766.

7. Testing

The Selenium suite previously completed with 3 tests, 0 failures, 0 errors, and 0 skipped tests and was integrated into the Jenkins build/test workflow.

8. Docker Deployment

The application is packaged with Eclipse Temurin 21 JRE. The container uses application port 8765 and host port 8766.

9. Puppet Reliability

Puppet 8.10.0 maintains C:/DeliveryPromise. Idempotency and recovery after configuration drift were both demonstrated.

10. Git Collaboration

Feature branches and pull requests were used throughout the project, including conflict resolution and the Puppet configuration merge.

11. Viva Questions

Question	Short answer
What is CI?	Automatically building/testing integrated changes.
What is CD?	Automatically deploying a successfully built release.
Why Docker?	Consistent, isolated and reproducible deployment.
Why port 8766?	Host 8765 was already occupied; container 8765 is mapped to host 8766.
What is Puppet?	Configuration management that enforces a desired state.
What is idempotency?	Repeated application reaches the same state without unnecessary changes.
Why Selenium?	Automated browser-level functional testing.
Why Jenkinsfile?	The CI/CD pipeline is stored and versioned as code.
What does Maven do?	Manages Java dependencies and build/test/package lifecycle.
What happens on failure?	Jenkins marks the stage/build failed and its logs are inspected.

12. Conclusion

The project demonstrates a complete DevOps chain: Git/GitHub for source control, Maven for Java build/test, Selenium for functional testing, Jenkins for CI/CD, Docker for containerized deployment, and Puppet for repeatable configuration management and recovery.

Task 15 — Actual Project Screenshots

Screenshots captured during the implementation are included below as evidence. These are the actual project screenshots, not recreated diagrams.

Figure 1. Final end-to-end Jenkins/Docker release evidence.

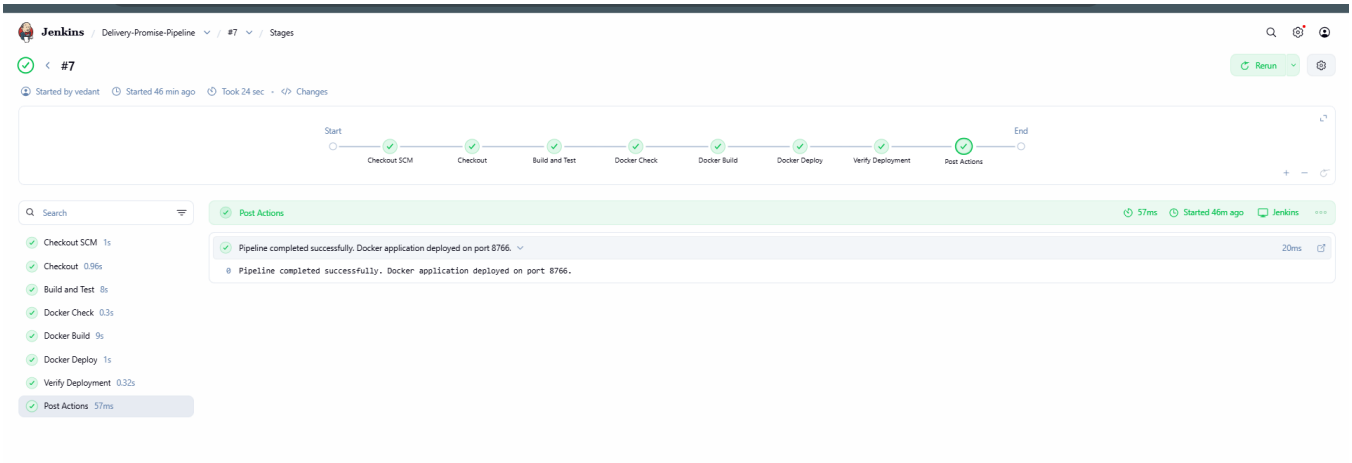


Figure 2. Final browser verification of the deployed application.

